

JEE MAIN REVISION MASTER: STRAIGHT LINES AND PAIR OF STRAIGHT LINES SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 171 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Distance, Slopes & Forms of Straight Line	100 PYQs	Straight Lines
Very High	Pair of Straight Lines & Angle Bisectors	74 PYQs	Homogeneous Lines

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Distance, Slopes & Forms of Straight Line	$d = \frac{ a x_1 + b y_1 + c }{\sqrt{a^2 + b^2}}$, $d_{\text{parallel}} = \frac{ c_1 - c_2 }{\sqrt{a^2 + b^2}}$, $\frac{x - x_1}{a} = \frac{y - y_1}{b} = -\frac{ax_1 + by_1 + c}{a^2 + b^2}$
Pair of Straight Lines & Angle Bisectors	$\tan \theta = \left \frac{2\sqrt{h^2 - ab}}{a + b} \right $, $\text{Perpendicular} \iff a + b = 0$, $\text{Coincident} \iff h^2 = ab$, $\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Distance, Slopes & Forms of Straight Line (Very High Yield — 100 PYQs)

Core Principle: Linear equations, distance formulas, family of lines $L_1 + \lambda L_2 = 0$, and coordinates of reflections (images) in lines.

■ **Caution / JEE Pitfall:** For distance between parallel lines, ensure coefficients of x and y are MADE IDENTICAL before subtracting $c_1 - c_2$.

■ **Speed Trick:** Image of point (x_1, y_1) in line $ax + by + c = 0$ is given by $(x - x_1 / a) = (y - y_1 / b) = -2(ax_1 + by_1 + c) / (a^2 + b^2)$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "If a straight line drawn through the point of intersection of the lines $4x+3y-1=0$ and $3x+4y-1=0$, meets the co-ordinate axes at the poin..." [Ans: N/A]
- [2026 Morning Shift] Q#2 (MCQ): "In an equilateral triangle PQR, let the vertex P be at (3,5) and the side QR be along the line $x+y=4$. If the orthocentre of the ..." [Ans: D]
- [2026 Morning Shift] Q#2 (MCQ): "Let ABC be an equilateral triangle with orthocenter at the origin and the side BC on the line $x+2\sqrt{2}y=4$. If the co-ordinates of the ..." [Ans: N/A]

3.2 Pair of Straight Lines & Angle Bisectors (Very High Yield — 74 PYQs)

Core Principle: Joint equations of lines through origin, condition of perpendicularity ($a + b = 0$), and joint equation of angle bisectors.

■ **Caution / JEE Pitfall:** If $a + b = 0$, the lines represented by $ax^2 + 2hxy + by^2 = 0$ are mutually perpendicular regardless of value of h .

■ **Speed Trick:** General second degree equation represents pair of lines if $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "If a straight line drawn through the point of intersection of the lines $4x+3y-1=0$ and $3x+4y-1=0$, meets the co-ordinate axes at the poin..." **[Ans: N/A]**
- **[2026 Morning Shift] Q#1 (MCQ):** "Let the mid points of the sides of a triangle ABC be $((5/2), 7)$, $((5/2), 3)$ and $(4, 5)$. If its incentre is $(\blacksquare, k)$, then $\$...$ " **[Ans: C]**
- **[2026 Evening Shift] Q#1 (MCQ):** "Let $A(1, 2)$ and $C(-3, -6)$ be two diagonally opposite vertices of a rhombus, whose sides AD and BC are parallel to the line $7x-y=14$. If ..." **[Ans: B]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 2nd April Morning Shift (Q#1)

Let the mid points of the sides of a triangle ABC be $((5/2), 7)$, $((5/2), 3)$ and $(4, 5)$. If its incentre is $(\blacksquare, k)$, then $3\blacksquare+k$ is equal to :

- (A) 11
- (B) 12
- (C) 13 **[Correct]**
- (D) 14

Correct Answer: C

Solution Analysis: Let the midpoints of the sides of triangle ABC be $D((5/2), 7)$, $E((5/2), 3)$, $F(4, 5)$. We have to find the incentre $(\blacksquare, k)$ of triangle ABC. Step 1: Use the fact about the medial triangle. The triangle formed by joining the midpoints of the sides of a triangle is called the medial triangle. If the midpoints are known, then the original triangle can be reconstructed using: $A=E+F-D$, $B=F+D-E$, $C=D+E-F$. Now calculate each vertex. Vertex A $A=((5/2), 3)+(4, 5)-((5/2), 7)$
 $A=(4, 1)$ Vertex B $B=(4, 5)+((5/2), 7)-((5/2), 3)$ $B=(4, 9)$ Vertex C $C=((5/2), 7)+((5/2), 3)-(4, 5)$
 $C=(5, 5)$

Benchmark Problem 2: JEE Main 2026 (Online) 5th April Morning Shift (Q#2)

In an equilateral triangle PQR, let the vertex P be at (3,5) and the side QR be along the line $x+y=4$. If the orthocentre of the triangle PQR is (α, β) , then $9(\alpha+\beta)$ is equal to:

- (A) 16
- (B) 27
- (C) 36
- (D) 48 [Correct]

Correct Answer: D

Solution Analysis: Since QR lies on the line $x+y=4$, and $P=(3,5)$, first find the altitude from P to the side QR. In an equilateral triangle, all altitudes, medians and angle bisectors coincide. Also, the orthocentre lies on the altitude from each vertex and divides the altitude in the ratio 2:1 from the vertex. So let us proceed step by step. The line QR has equation $x+y-4=0$. Distance of point P(3,5) from this line is $d = \frac{|3+5-4|}{\sqrt{1^2+1^2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2}$. This is the altitude of the equilateral triangle from vertex P. Now, the foot of the perpendicular from P